

A/B Testing Product Experimentation Analytics

Tech Stack: MySQL, Python

Project Overview

Analyzed impact of CTA button redesign using controlled A/B testing experiment on 40,000 users and 80,000 sessions.

Business Problem

Evaluate whether redesigned CTA button improves engagement and conversions before global rollout.

Dataset

40,000 users | 80,000 sessions | 13,600 clicks | 7,600 conversions | A/B cohorts (20k each)

Schema

users, experiment_group, sessions, clicks, conversions, events

Metrics Analyzed

1. CTR: A=30%, B=38%
2. Conversion Rate: A=18%, B=20%
3. Conversion Uplift: 11.11%
4. Avg Session Duration: A=145 sec, B=172 sec
5. Bounce Rate: Group B lower than Group A

SQL Analysis

Used joins, CTEs, aggregate functions, behavioral analysis logic, cohort comparison queries.

Key Insights

Treatment variant improved engagement, increased click-through behavior, reduced bounce rate, and improved conversion performance.

Final Recommendation

Roll out Variant B to all users based on statistically meaningful performance improvement.

Resume Impact

Demonstrates product analytics thinking, experiment design, SQL proficiency, and user behavior analysis similar to real product companies.